

Laboratório 04 - Importação de Dados Retangulares - Excel

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Q1. Carregar pacotes

```
library(readxl)
library(tidyverse)
```

Q2. Caminho e verificação

```
fname <- "datasets/WDIEXCEL.xlsx"
file.exists(fname)
```

Q3. Ler Country

```
countries <- read_excel(fname, sheet = "Country")
```

Q4. Manter cinco países

```
countries <- countries %>%
  filter(`Short Name` %in% c("Brazil", "Argentina", "Chile", "Mexico", "United States"))
```

Q5. Ler Data

```
WDI <- read_excel(fname, sheet = "Data")
```

Q6. Selecionar colunas

```
WDI <- WDI %>%
  select(`Country Code`, `Indicator Code`, `1960`:`2018`)
```

Q7. Formato tidy

```
WDI <- WDI %>%
  pivot_longer(cols = `1960`:`2018`, names_to = "YEAR", values_to = "VALUE")
```

Q8. YEAR inteiro

```
WDI <- WDI %>% mutate(YEAR = as.integer(YEAR))
is.integer(WDI$YEAR)
```

Q9. Filtrar países e selecionar variáveis

```
WDI <- WDI %>%
  filter(`Country Code` %in% countries$`Country Code`) %>%
  select(`Country Code`, `Short Name`, YEAR, `SP.DYN.CBRT.IN`)
```

Q10. Ordenar

```
WDI <- WDI %>% arrange(`Short Name`, YEAR)
```

Q11. Gráfico

```
ggplot(WDI, aes(x = YEAR, y = `SP.DYN.CBRT.IN`, color = `Short Name`)) +
  geom_line() +
  labs(x = "YEAR", y = "SP.DYN.CBRT.IN", color = "País")
```

Data e hora da compilação

```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```